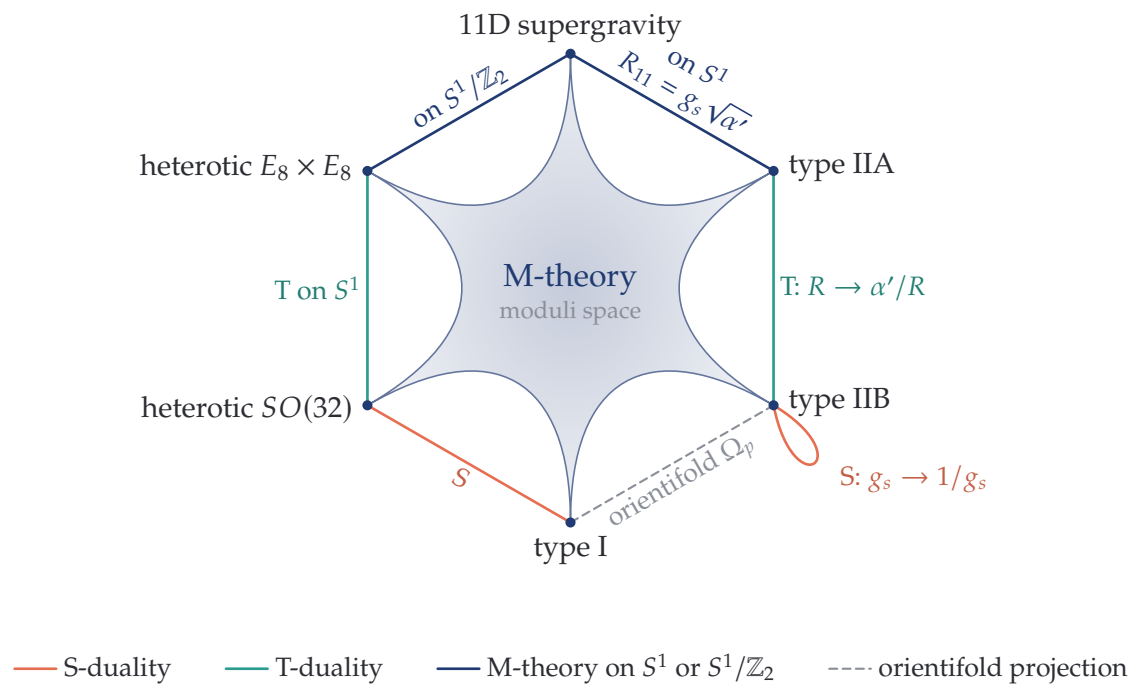

11D supergravity
on $S^1/\mathbb{Z}_2$
$R_{11} = g_s\sqrt{\alpha'}$ on S^1
heterotic $E_8 \times E_8$
type IIA
T on S^1
M-theory
moduli space
T: $R \to \alpha'/R$
heterotic $SO(32)$
type IIB
S
orientifold Ω_p
S: $g_s \to 1/g_s$
type I
S-duality
T-duality
M-theory on S^1 or $S^1/\mathbb{Z}_2$
orientifold projection